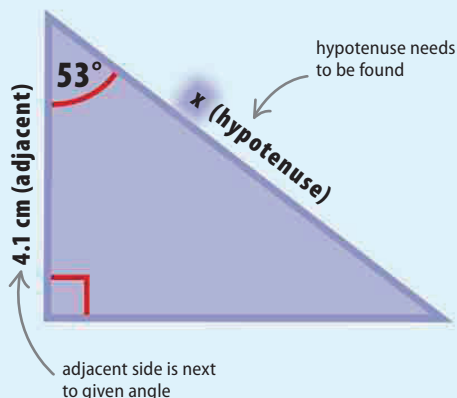


## Using the cosine formula

In this right triangle, one angle other than the right angle is known, as is the length of the side adjacent to it. The hypotenuse is the missing side that needs to be found.



**Choose the right formula**—because the side adjacent to the angle is known and the value of the hypotenuse is missing, use the cosine formula.

**Substitute** the known values into the formula.

**Rearrange** to make  $x$  the subject of the equation—first multiply both sides by  $x$ .

**Divide both sides** by  $\cos 53^\circ$  to make  $x$  the subject of the equation.

**Use a calculator** to find the value of  $\cos 53^\circ$ —press the cos button then enter 53.

**Round the answer** to a suitable size.

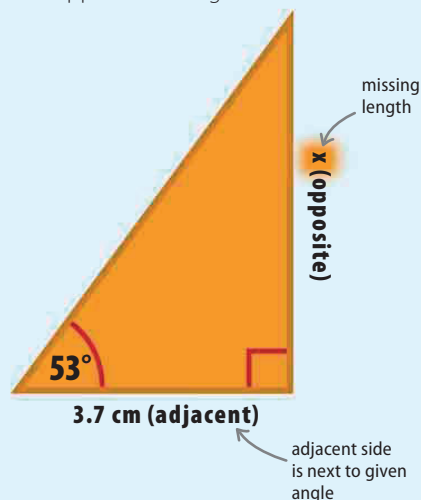
$$\begin{aligned} \cos A &= \frac{\text{adjacent}}{\text{hypotenuse}} \\ \cos 53^\circ &= \frac{4.1}{x} \\ \cos 53^\circ \times x &= 4.1 \\ x &= \frac{4.1}{\cos 53^\circ} \\ x &= \frac{4.1}{0.6018} \\ x &= 6.81 \text{ cm} \end{aligned}$$

Annotations for the cosine formula steps:

- For  $\cos 53^\circ = \frac{4.1}{x}$ : "this side has also been multiplied by  $x$ , leaving 4.1 on its own"
- For  $\cos 53^\circ \times x = 4.1$ : "this side has been multiplied by  $x$ "
- For  $x = \frac{4.1}{\cos 53^\circ}$ : "this side has also been divided by  $\cos 53^\circ$ "
- For  $x = \frac{4.1}{0.6018}$ : "value of  $\cos 53^\circ$  is rounded to 4 decimal places"
- For  $x = 6.81 \text{ cm}$ : "answer is rounded to 2 decimal places"

## Using the tangent formula

In this right triangle, one angle other than the right angle is known, as is the length of the side adjacent to it. Find the length of the side opposite the angle.



**Choose the right formula**—since the side adjacent to the angle given are known and the opposite side is sought, use the tangent formula.

**Substitute** the known values into the tangent formula.

**Rearrange** to make  $x$  the subject by multiplying both sides by 3.7.

**Use a calculator** to find the value of  $\tan 53^\circ$ —press the tan button then enter 53.

**Round the answer** to a suitable size.

$$\begin{aligned} \tan A &= \frac{\text{opposite}}{\text{adjacent}} \\ \tan 53^\circ &= \frac{x}{3.7} \\ x &= \tan 53^\circ \times 3.7 \\ x &= 1.3270 \times 3.7 \\ x &= 4.91 \text{ cm} \end{aligned}$$

Annotations for the tangent formula steps:

- For  $\tan 53^\circ = \frac{x}{3.7}$ : "this side has also been multiplied by 3.7"
- For  $x = \tan 53^\circ \times 3.7$ : "show the unknown first"
- For  $x = 1.3270 \times 3.7$ : "this side has been multiplied by 3.7 to isolate  $x$ "
- For  $x = 1.3270 \times 3.7$ : "value of  $\tan 53^\circ$  is rounded to 4 decimal places"
- For  $x = 4.91 \text{ cm}$ : "the answer is rounded to 2 decimal places"